From Spectra to Topology

The Hodge–Dijkgraaf–Witten bridge: harmonic qubits, holonomy,
the Choi bend, and finding geometry by minimizing eigenvector residuals

tessera — cobordism realizability programme

Abstract

This primer assembles the theory behind the realizability bridge. We are after a single statement of the *state–operation–cobordism correspondence*: a quantum operation U is realizable as a bulk cobordism W_{AB} whose value reproduces the transition amplitude $\langle\psi_A|U|\psi_B\rangle$. There are two roads to that value. The *spectral* road builds the geometry: states become harmonic forms of a combinatorial Hodge Laplacian, and we *synthesize* a bulk whose boundary Laplacian carries the (Choi-bent) operator as an eigenvector. The *topological* road is the Dijkgraaf–Witten state sum: a metric-free invariant computed from the cobordism's topology and a group cocycle. The bridge asks whether the two roads give the same number. We develop, in order: the combinatorial Hodge Laplacian and its harmonic spaces; the *Hodge qubit*; flat $\mathbb{Z}_2$ holonomy and the Dijkgraaf–Witten partition function; the crucial distinction between the b_1-dimensional harmonic space and the 2^{b_1}-dimensional holonomy space, and the preparation map between them; the Choi–Jamiołkowski "bend" that turns an operation into a state; and a whitepaper-style algorithm that recovers *topology* by driving an eigenvector residual to zero.

Contents

1 The realization map geo$(\cdot)$ **1**

2 Overview: two roads to $Z(W_{AB})$ **2**

3 The combinatorial Hodge Laplacian 3

4 Harmonic forms and the Hodge qubit 4

5 Holonomy, flat connections, and Dijkgraaf–Witten 4

6 Two countings, one preparation map 5

7 Choi–Jamiołkowski: operations as states (the bend) 5

8 Finding topology by minimizing eigenvector residuals 6

9 The bridge 6

1 The realization map geo$(\cdot)$

Informally, geo is the *realization* (synthesis) map that turns quantum data into geometry — the inverse of the usual "read a spectrum off a given complex" pipeline. It acts on two kinds of input.

Definition 1 (The realization map)**.** *On states.* For a normalized quantum state ψ, $\mathrm{geo}(\psi)$ is a closed oriented *weighted* simplicial surface Σ that carries ψ as a harmonic form: $\psi \in \ker L_1\big(\mathrm{geo}(\psi)\big)$. The amplitudes and relative phase of ψ are encoded as the edge weights and holonomies of that harmonic 1-form (§4). A single qubit realizes as the torus ($g = 1$, $b_1 = 2$); a g-qubit state as a genus-g surface.

On operations. For an operation $U : \mathcal{H}_B \to \mathcal{H}_A$, $\mathrm{geo}(U)$ is a cobordism (a manifold-with-boundary) whose interior weighted triangulation carries the Choi-bent operator $\mathrm{vec}(U)$ (§7) as a boundary harmonic; this is the bulk W_{AB} of the Overview.

Four properties pin down how geo behaves; they are what the later sections silently assume.

(G1) Boundary-compatibility. Restriction of an operation to its boundary states commutes with geo: $\partial\, \mathrm{geo}(U) = \overline{\mathrm{geo}(\psi_B)} \sqcup \mathrm{geo}(\psi_A)$, the outgoing geometry of ψ_A together with the orientation-reversed incoming geometry of ψ_B. (This is hypothesis (H2) of the Overview.) So the state-level and operation-level halves of Definition 1 agree on shared boundaries.

(G2) Metric-laden. geo outputs a *weighted* complex — edge weights and $U(1)$ phases, i.e. a metric and a connection, not just a triangulation. The spectral value Z_{spec} depends on these weights, in contrast to the Dijkgraaf–Witten road (§5), which discards them.

(G3) Defined implicitly, by spectral realization. There is no closed formula for geo. A complex W *is* $\mathrm{geo}(U)$ exactly when the eigenvector residual $r(W)$ of §8 vanishes — when the bent target is an eigenvector of the boundary Hodge Laplacian. The inverse problem of §8 *is* the constructive content of geo; the realizability oracle is the search that evaluates it.

(G4) Partial and non-unique. $\mathrm{geo}(U)$ need not exist: it is defined only on the *realizable* operations. For an obstructed U the residual floors above zero and $\mathrm{geo}(U)$ is undefined — the floor is the non-existence certificate (§8). When it does exist it is not unique: gauge/metric freedom and boundary-fixed Pachner moves yield many witnesses with $r(W) = 0$, and $\mathrm{geo}(U)$ denotes any one of them.

With geo in hand the three hypotheses of the next section read directly: (H1) $\mathrm{geo}(U)$ is a cobordism, (H2) its boundary is geo of the two states (G1), and (H3) its invariant is the amplitude. In the codebase geo on operations is realized by `EigenstateSynthesis` / `RealizabilityOracle.decide()` (§8) with the bend supplied by `ChoiJamiolkowski` (§7); the witness is returned as a `Spacetime` and `getBoundary()` recovers $\partial\, \mathrm{geo}(U)$.

2 Overview: two roads to $Z(W_{AB})$

Fix two boundary states ψ_A, ψ_B and an operation $U : \mathcal{H}_B \to \mathcal{H}_A$. The correspondence claims three things about a bulk W_{AB} built "from" U:

(H1) $W_{AB} = \mathrm{geo}(U)$ is a cobordism (a manifold-with-boundary);

(H2) $\partial W_{AB} = \overline{\mathrm{geo}(\psi_B)} \sqcup \mathrm{geo}(\psi_A)$ — its boundary is the (outgoing) geometry of ψ_A and the orientation-reversed (incoming) geometry of ψ_B;

(H3) $Z(W_{AB}) = \langle \psi_A | U | \psi_B \rangle$ — the cobordism's invariant equals the transition amplitude.

The subtlety lives in (H3), because "Z" has two meanings:

- **Spectral / geometric Z.** Realize W_{AB} by *synthesizing* it: pin its boundary, fill its interior (edge weights and topology) until the boundary Hodge–Laplacian eigenvector matches the bent operator. The value is read off the geometry; it depends on the metric (edge weights).
- **Topological Z (Dijkgraaf–Witten).** Compute the $\mathbb{Z}_2$ state sum from the triangulation and a cocycle alone — no edge weights. The value is a topological invariant.

The bridge is the claim that these coincide on W_{AB}. The rest of this primer builds the machinery on both sides and the map between their boundary data.

3 The combinatorial Hodge Laplacian

Let K be a (finite, oriented) simplicial complex with weighted edges. Write C^k for the space of $\mathbb{C}$-valued k-cochains (functions on oriented k-simplices, antisymmetric under reorientation). The *coboundary* $d_k : C^k \to C^{k+1}$ is the signed incidence operator,

$$(d_k\alpha)(\sigma) \;=\; \sum_{\tau\subset\sigma,\ \dim\tau=k} [\sigma:\tau]\,\alpha(\tau), \tag{1}$$

with $[\sigma:\tau]\in\{\pm1\}$ the relative orientation. The defining identity of a chain complex is $d_{k+1}d_k = 0$.

A choice of Hermitian inner products on each C^k (here induced by the edge weights, see Remark 1) gives adjoints $d_k^\dagger : C^{k+1}\to C^k$. The *Hodge Laplacian* in degree k is

$$L_k \;=\; \underbrace{d_{k-1}\,d_{k-1}^\dagger}_{\text{“down”}} + \underbrace{d_k^\dagger\,d_k}_{\text{“up”}}, \qquad L_k = L_k^\dagger \succeq 0. \tag{2}$$

Remark 1 (The magnitude convention at $k=0$). At $k=0$ the down-term vanishes and $L_0 = d_0^\dagger d_0$ is the graph Laplacian. With a *Hermitian* weight $A_{ij} = w_{ij}\,e^{i\phi_{ij}}$ on the edge $i\to j$ (so $A_{ji} = \overline{A_{ij}}$) and the *magnitude* degree $D_{ii} = \sum_j |w_{ij}|$, the project uses

$$L_0 \;=\; D - A, \qquad D = \operatorname{diag}(D_{ii}), \tag{3}$$

a Hermitian operator whose evolution e^{-iL_0t} is unitary. The phase ϕ_{ij} is a $U(1)$ connection living on the edge; its loop sums are gauge-invariant fluxes. This is exactly `cobordism::HodgeLaplacian` (the class is degree-parameterized: $k=0$ from edge weights, $k\ge1$ from the boundary matrices of `ChainComplex`).

The central fact is the discrete Hodge theorem.

Definition 2 (Harmonic cochains). $\alpha\in C^k$ is *harmonic* iff $L_k\alpha=0$; equivalently $d_k\alpha=0$ and $d_{k-1}^\dagger\alpha = 0$ (closed and co-closed).

Theorem 1 (Discrete Hodge decomposition). $C^k = \operatorname{im} d_{k-1} \,\oplus\, \ker L_k \,\oplus\, \operatorname{im} d_k^\dagger$*, and the harmonic space is isomorphic to cohomology,*

$$\ker L_k \;\cong\; H^k(K;\mathbb{C}), \qquad \boxed{\dim\ker L_k \;=\; b_k} \tag{4}$$

the k-th Betti number. Harmonic representatives are the unique minimum-norm elements of their cohomology classes.

So the *kernel* of L_k is a coordinate-free handle on topology: its dimension counts k-dimensional “holes,” and its elements are canonical (metric-orthogonalized) representatives of those classes. The eigenvectors of L_k with small nonzero eigenvalue are the “almost-harmonic” modes; the spectrum interpolates between geometry (large eigenvalues, local) and topology (the kernel, global).

4 Harmonic forms and the Hodge qubit

Take Σ a closed oriented surface of genus g. Then $b_1(\Sigma) = 2g$, so $\dim\ker L_1(\Sigma) = 2g$: there are exactly $2g$ independent harmonic 1-forms, one "dual" pair per handle.

Definition 3 (Hodge qubit)**.** For the torus T^2 ($g = 1$) we have $\dim\ker L_1(T^2) = 2$: a two-dimensional complex vector space. We call it the *Hodge qubit.* A unit harmonic form

$$\psi = \alpha\,\omega_x + \beta\,\omega_y \in \ker L_1(T^2), \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{5}$$

written in the basis of the two harmonic generators ω_x, ω_y (the discrete analogues of dx, dy around the two cycles), *is* the qubit state $\alpha|0\rangle + \beta|1\rangle$.

The amplitudes and relative phase of a quantum state are encoded as the weights and holonomies of a harmonic 1-form on a surface. Boundary states for the cobordism are chosen as $\psi_A \in \ker L_1(\Sigma_A)$, $\psi_B \in \ker L_1(\Sigma_B)$. (The $k = 0$ analogue — a node-amplitude qubit $\ker L_0$ on a small graph — is the reduced setting used for the first realizability oracle; the surface/k=1 version is the one that matches Dijkgraaf–Witten, §5–§6.)

5 Holonomy, flat connections, and Dijkgraaf–Witten

Now switch from continuous harmonic forms to discrete gauge fields. A *flat* $\mathbb{Z}_2$ *connection* on a complex assigns $g_e \in \mathbb{Z}_2 = \{0, 1\}$ to each edge so that the holonomy around every triangle is trivial; modulo gauge transformations these are classified by

$$H^1(\Sigma; \mathbb{Z}_2) \;=\; \mathrm{Hom}\big(H_1(\Sigma), \mathbb{Z}_2\big) \;\cong\; (\mathbb{Z}_2)^{\,b_1(\Sigma)}. \tag{6}$$

A class is fixed by its *holonomy* ± 1 around each of the b_1 independent cycles, so there are $2^{\,b_1}$ of them.

Definition 4 (Dijkgraaf–Witten boundary space)**.** A boundary surface Σ carries the Hilbert space

$$Z(\Sigma) \;=\; \mathbb{C}\big[\,H^1(\Sigma; \mathbb{Z}_2)\,\big], \qquad \dim Z(\Sigma) = 2^{\,b_1(\Sigma)}, \tag{7}$$

spanned by the holonomy classes (one basis vector per flat connection).

For a finite group G (here $\mathbb{Z}_2$) and a group n-cocycle $\omega \in H^n(BG; U(1))$, the *Dijkgraaf–Witten state sum* of a closed n-manifold W is

$$Z(W) \;=\; \frac{1}{|G|^{\,|V|}} \sum_{\substack{g:\,\text{edges}\to G\\ \text{flat}}} \;\prod_{n\text{-simplices } t} \omega(g_t)^{\,\varepsilon_t}, \tag{8}$$

a number; $\varepsilon_t = \pm 1$ is the orientation of the top simplex t. For $n = 3$ and $G = \mathbb{Z}_2$, ω is one of two 3-cocycles — the trivial one ($\omega \equiv 1$) and the nontrivial *sign* class ($\omega = (-1)^{abc}$). For a cobordism W with boundary one holds the boundary connection fixed and sums the same product over interior flat fields; the result is a vector in $Z(\partial W)$, equivalently (for two boundary components) a linear map

$$Z(W) : Z(\Sigma_B) \longrightarrow Z(\Sigma_A), \qquad \langle \psi_A | Z(W) | \psi_B \rangle \;=\; \sum_{a,b} \overline{\psi_A[a]}\; Z(W)_{ab}\; \psi_B[b]. \tag{9}$$

For the trivial cylinder $\Sigma \times [0, T]$ this map is the identity on $Z(\Sigma)$. In the codebase this is `cobordism::DijkgraafWitten`: `partitionFunction()` (closed), `map()` / `boundaryVector()` (with boundary), and `amplitude(`ψ_A, ψ_B`)` for the sandwiched value above.

6 Two countings, one preparation map

We now have *two* boundary spaces attached to the same surface Σ, and they are *not* the same:

	Hodge qubit	Dijkgraaf–Witten space
object	harmonic 1-forms $\ker L_1(\Sigma)$	holonomy classes $\mathbb{C}[H^1(\Sigma;\mathbb{Z}_2)]$
dimension	b_1 (continuous, over $\mathbb{C}$)	$2^{\,b_1}$ (discrete count)
torus T^2	$\mathbb{C}^2$ (one qubit)	$\mathbb{C}^4$
role	the spectral state	the state-sum basis

Remark 2 (Keep the countings distinct). b_1 versus $2^{\,b_1}$: continuous harmonic forms give the *qubit dimension*; flat connections *index the state sum.* They coincide only trivially. This is stated as a standing convention in the experiment charter (§5.2), and the Dijkgraaf–Witten class deliberately keeps the "b_1 qubit" separate from the "$2^{\,b_1}$ flat-connection count."

The two sides meet through a *preparation map*

$$\mathsf{prep}:\ \ker L_1(\Sigma)\ \longrightarrow\ \mathbb{C}[H^1(\Sigma;\mathbb{Z}_2)],\qquad b_1\mapsto 2^{\,b_1},\tag{10}$$

which encodes a harmonic (spectral) boundary state as an amplitude over holonomy classes; `DijkgraafWitten.amplitude()` already consumes states in the flat-connection basis that have been prepared this way. Making prep and its readout first-class and tested is the foundational step of the bridge work — it is the only place the spectral and topological descriptions of the *boundary* must be reconciled. (Everything "interesting" then happens in the *bulk* invariant.)

7 Choi–Jamiołkowski: operations as states (the bend)

To feed an operation into a geometry we must first turn it into a state. The vehicle is the Choi–Jamiołkowski isomorphism. Define the *vectorization* of $U=\sum_{ij}U_{ij}\,|i\rangle\langle j|:\mathcal{H}_B\to\mathcal{H}_A$ by stacking rows,

$$\mathrm{vec}(U)\ =\ \sum_{ij}U_{ij}\,|i\rangle\otimes|j\rangle\in\mathcal{H}_A\otimes\mathcal{H}_B,\qquad \mathrm{vec}\big(|a\rangle\langle b|\big)\ =\ |a\rangle\otimes\overline{|b\rangle}.\tag{11}$$

The operator becomes a (generally entangled) bipartite state — the *Choi state.* Two facts make it the right bend.

Theorem 2 (Hilbert–Schmidt / Choi identity). *For the rank-one transition operator* $U_T=|\psi_A\rangle\langle\psi_B|$,

$$\langle\psi_A|U|\psi_B\rangle\ =\ \mathrm{Tr}\big(U_T^\dagger U\big)\ =\ \langle\mathrm{vec}(U_T),\mathrm{vec}(U)\rangle.\tag{12}$$

That is: the physical amplitude $\langle\psi_A|U|\psi_B\rangle$ equals an *inner product of Choi states.* So if a cobordism faithfully carries $\mathrm{vec}(U)$ on its output boundary, contracting that boundary state against the test functional $\mathrm{vec}(U_T)$ returns the amplitude — which is exactly how the spectral side reads off "Z." (This is `quantum::ChoiJamiolkowski`: `vectorize`, `transitionOperator`, `transitionAmplitude`.)

Theorem 3 (Rank = Schmidt rank = connectivity). *The Schmidt rank of* $\mathrm{vec}(U)$ *equals* $\mathrm{rank}(U)$ *(the number of nonzero singular values of U). A separable Choi state (*rank 1*) corresponds to a rank-one* $U_T=|\psi_A\rangle\langle\psi_B|$ *and a* disconnected *cup/cap geometry; higher Schmidt rank is genuine entanglement and a* connected *bulk.*

8 Finding topology by minimizing eigenvector residuals

We can now state the inverse problem precisely. Given a target boundary state ψ (the bent operator, $\psi = \mathrm{vec}(U)/\|\mathrm{vec}(U)\|$), *find a complex W whose Hodge Laplacian has ψ as an eigenvector.* The eigenvalue is a free output, so we use the eigenvalue-agnostic *residual*

$$\boxed{r(W) \;=\; \big\|\, \big(I - \psi\psi^\dagger\big)\, L(W)\,\psi\,\big\|^2 \;=\; \big\|\, L\psi - (\psi^\dagger L\psi)\,\psi\,\big\|^2.} \tag{13}$$

$r(W)$ is the squared norm of the component of $L\psi$ orthogonal to ψ; it vanishes *iff* $L\psi = \lambda\psi$ with $\lambda = \psi^\dagger L\psi$ the Rayleigh quotient. Minimizing r over the degrees of freedom of W is thus a search for a geometry that *realizes* ψ spectrally.

Fixed-boundary fill. For realizability the boundary is *pinned*: ∂W is the synthesized $\mathrm{geo}(\psi_A)$, $\mathrm{geo}(\psi_B)$ and the output surface, held fixed. The free parameters θ are the *interior* edge weights/phases and — through boundary-fixed Pachner "add" moves (coning) — the interior *topology*. This is the decisive twist: with the ends fixed the parameters do *not* outrun the constraints, so realizability becomes a genuine test rather than a foregone conclusion.

Verdict and certificate. A target is **realizable** iff r can be driven below ϵ; it is **obstructed** iff r *floors* at a positive value — a residual lower bound under the fixed-boundary constraint. The floor is the non-existence *certificate* (the analogue of the two-vertex floor $w_{\min}^2(|c_0|^2 - |c_1|^2)^2$), obtained without exhausting triangulations.

Algorithm. $\textsc{DecideRealizability}(U,\, d_A,\, d_B;\ \epsilon,\, R,\, c_{\max})$

Input: operation $U : \mathcal{H}_B \to \mathcal{H}_A$; tolerance ϵ; restart count R; cone budget $c_{\max}$.
Output: REALIZABLE with witness W, or OBSTRUCTED with floor f.

```
ψ ← vec(U) / ‖vec(U)‖                                  // bend: Choi–Jamiołkowski (§7)
W ← seed bulk;  Pin(∂W)                                 // fixed ends; only interior edges are writable
f ← +∞
for  c = 0, 1, ..., c_max  do
    define r_W(θ) = ‖(I − ψψ†) L(W; θ) ψ‖²               // residual over interior params θ
    (θ*, r*) ← MultiRestartLM(r_W, R)                   // Levenberg–Marquardt ×R seeds
    if  r* < ε  then return  REALIZABLE, W(θ*)           // witness carries ψ as an eigenvector
    f ← min(f, r*)
    W ← ConeInterior(W)                                 // boundary-fixed Pachner add: +1 interior vertex
return  OBSTRUCTED, floor  f                            // f bounded away from 0 certifies non-existence
```

In the codebase: `ChoiJamiolkowski` does the bend; `EigenstateSynthesis` evaluates r_W and the fixed-boundary fill; `LevenbergMarquardt` is the multi-restart solver; `RealizabilityOracle.decide()` is the loop above; the witness W is returned as a `Spacetime`, and `getBoundary()` recovers ∂W.

9 The bridge

Putting the pieces together on a single synthesized W_{AB}:

- the spectral road (§8) produces W_{AB} whose boundary harmonic carries $\mathrm{vec}(U)$, so by Theorem 2 the *spectral* $Z_{\mathrm{spec}}(W_{AB}) = \langle\psi_A|U|\psi_B\rangle$;

- the topological road (§5) computes $Z_{\mathrm{DW}}(W_{AB}) = \langle \psi_A | Z(W_{AB}) | \psi_B \rangle$ from the triangulation and cocycle alone, with boundary states obtained through the preparation map of §6.

The bridge is the assertion

$$Z_{\mathrm{DW}}(W_{AB}) \stackrel{?}{=} \langle \psi_A | U | \psi_B \rangle \stackrel{?}{=} Z_{\mathrm{spec}}(W_{AB}). \tag{14}$$

Remark 3 (A discrete shadow). The $\mathbb{Z}_2$ Dijkgraaf–Witten functor sends cobordisms to a *discrete* family of maps (the Frobenius algebra $\mathbb{C}[\mathbb{Z}_2]$ generated by pair-of-pants, cap/cup, cylinder, twisted by the cocycle). A generic U is *not* Z_{DW} of any $\mathbb{Z}_2$ cobordism. The honest expectation, then, is that the bridge holds on the discrete *DW-representable* subset, and that the spectral oracle — which realizes a *continuum* of operators — strictly extends the topological theory. In that picture the Dijkgraaf–Witten invariant is the *quantized shadow* of the spectral Z: equal where both are defined, with the geometry seeing strictly more than the finite-group state sum.

Where this lives. `HodgeLaplacian` (§3–§4), `DijkgraafWitten` (§5–§6), `ChoiJamiolkowski` (§7), and `RealizabilityOracle/EigenstateSynthesis/LevenbergMarquardt` (§8) are in place; the preparation map of §6, the $k{=}1$ 3-manifold synthesis, and the agreement experiment of §9 are the three steps of the bridge milestone.